

Brain Corp Technical Operations Code Challenge

Pokémon API

Pokémon are little creatures with a TV show, a couple of cartoon movies, a few video games and most recently, a public RESTful API (<https://pokeapi.co/>) with all of their information. In techops, we care a lot about APIs so we're asking you to build a simple CLI tool to interact with one.

Among other things, each Pokémon has the following data:

- `name` - the name of the Pokémon
- `id` - the pokedex number of the Pokémon.
- `moves` - the attack/defense moves of the Pokémon.

For example, `Pikachu` has the pokedex number `25`, and knows the move `skull-bash`.

If you have `curl` installed on your machine, run the following to verify and see more information about `pikachu`:

```
curl -s https://pokeapi.co/api/v2/pokemon/pikachu
```

Prerequisites

You will need at least the following things properly installed on your computer.

- `python3`
- `pip3`

Code

To submit your solution, create a zip of your code. It should include:

- `README.md` - a markdown document with installation and running instructions and anything else that will help the user of your code.
- `pokemon.py` - the command line tool. We don't mind if it's an executable script or if it needs to be run as `python pokemon.py` (but see above about needing clear documentation!).
- Any other relevant Python files needed to make the CLI run. You do *not* need to provide any third-party code that you are relying on (or a Python runtime and so on).

Requirements

This is a simple CLI using the RESTful Pokémon API. This task is not meant to trick you. Here's the functionality we'd like you to implement:

Basic Pokémon Lookup

To start, we'd like to get some basic information on a given Pokémon. Specifically:

- Look up Pokémon by name, print out moves (alphabetical order).
- Look up Pokémon by pokedex number, print out moves (alphabetical order).
- Errors should be handled gracefully, with appropriate messages to the user.

Top 10 Most Common Moves (for the first 100 Pokémon)

As you look at the Pokémon data, you'll see that each of them have "moves". For example, `vine-whip`. We are interested which moves are most common. To be specific, by "most common" we mean which moves are associated with the greatest number of Pokémon.

For example, if we consider just the Pokémon `pikachu`, `jigglypuff` and `charizard`, all three have access to the `body-slam` move, but only `pikachu` and `jigglypuff` get to use `zap-cannon`. So we would consider the `body-slam` move to be more popular than `zap-cannon`.

We used to ask for the top 10 most common moves across all Pokémon. But there are now lots of Pokémon and we don't want you to have to ping the API hundreds of times to get an answer. So instead we want:

The top 10 most common moves *that begin with the letter 's'* across the *first 100* Pokémon (by id). We'd like the count of the moves as well as their names.

If there's a tie for the tenth position, you may handle this in any way you please.

Other Considerations

You are not required to only use what is listed in the `Requirements` section to get the job done. Use whatever libraries you deem necessary. But we *do* want it to be a Python 3-based solution.

We're looking not just for the code to work, but how you implemented it, documented it, how easy it is for someone else to run it, and how maintainable it would be in the future.

You may use Google, Stack Overflow, etc. but your solution must represent your own work. Use comments to give credit if you borrow anything from another source.

This test is not designed to fool you. If you feel there are any ambiguities, then the README is a great place to explain any design or implementation decisions you made based on interpreting our requests!

Also, please do not share this problem with anybody.